

Week 5: Cross-Encoder Reranking

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1. Objective

W4 showed that, on the qrels-anchored sampled corpus, dense retrieval already pushes Recall@100 to 0.9946. Recall is no longer where the gain comes from. W5 asks the next question:

Once recall has saturated, can a cross-encoder reranker still improve the *local ordering* — i.e. push the relevant passage from rank 3 to rank 1?

Pipeline:

dense top-100 → cross-encoder rerank → evaluate

We deliberately keep the first stage unchanged from W4 (no re-encoding, no new index) so the delta is attributable to the reranker alone.

2. Setup

- **First-stage retriever:** W4 dense run (outputs/week04_dense/run.tsv), truncated to top-100 per query.
- **Reranker:** cross-encoder/ms-marco-MiniLM-L-6-v2 — the standard MS MARCO cross-encoder baseline (small enough to run on CPU).
- **Eval set:** 1000 dev/small queries with ≥ 1 qrel landing in the W4 sampled pool.
- **Evaluation:** same sample-restricted qrels as W4, so the dense numbers reported in column 1 below are directly comparable to the W4 dense numbers.

3. Why top-100?

A cross-encoder is $O(K)$ forward passes per query — every (query, passage) pair goes through the model. $K=1000$ would be $\sim 10\times$ the cost of $K=100$ with diminishing returns once recall is already near ceiling. The standard depth in the MS MARCO reranking literature is 100, which is what we use here.

4. Results

Metric	Dense (top-100)	Dense + CE rerank	Δ (rerank – dense)
MRR@10	0.8846	0.9282	+0.0435
nDCG@10	0.9014	0.9412	+0.0398
Recall@100	0.9915	0.9915	+0.0000

Recall@100 is unchanged by construction: the reranker only re-orders the top-100 candidates the dense retriever already returned. The interesting columns are MRR@10 and nDCG@10 — both measure where in the ranked list the relevant passage actually lands.

5. Runtime

Quantity	Value
Rerank depth (K)	100
Pairs scored	100000
Wall-clock (rerank)	3493.4 s
Throughput	0.29 q/s, 29 pairs/s
Peak RSS	3889 MiB
Text resolution	11.8 s
Batch size	128
Max length	512

The cost is **linear in K**: doubling the rerank depth roughly doubles the wall-clock. Queries-per-second scales inversely.

6. Qualitative examples (before vs after)

Sampled queries where the cross-encoder moved the relevant passage:

195440 — gnosticism heresy definition

- Dense first-relevant rank: 1
- Rerank first-relevant rank: 1
- Relevant doc id(s): 7364279

414799 — is it windy in arizona

- Dense first-relevant rank: 1
- Rerank first-relevant rank: 1
- Relevant doc id(s): 7587115

448123 — meaning of last name, harrison

- Dense first-relevant rank: 1
- Rerank first-relevant rank: 1
- Relevant doc id(s): 7582932

688644 — what is a learning objective for teaching toddlers about weather

- Dense first-relevant rank: 1
- Rerank first-relevant rank: 1
- Relevant doc id(s): 7573780

877845 — what metal to keep boat compass away from

- Dense first-relevant rank: 1
- Rerank first-relevant rank: 1
- Relevant doc id(s): 7849787

7. Discussion

- **MRR@10**: cross-encoder improves the metric by 0.0435 (dense 0.8846 → rerank 0.9282).
- **nDCG@10**: cross-encoder improves the metric by 0.0398 (dense 0.9014 → rerank 0.9412).
- **Recall@100** is unchanged (0.9915) — by construction, the reranker only re-orders the top-K dense candidates and cannot recover docs the dense retriever missed.
- In the 20 sampled examples, the relevant passage was promoted in 0 and demoted in 0 (remaining cases unchanged or both >10).

8. Limitations

- **No fine-tuning**: cross-encoder/ms-marco-MiniLM-L-6-v2 is the published MS MARCO baseline weights, used as-is. A larger or domain-tuned cross-encoder would shift the numbers further.
- **Sampled corpus**: the first-stage W4 dense run is on the qrels- anchored 50k sample, not the full 8.8M corpus. The reranker delta (+0.0435 on MRR@10) should transfer qualitatively to the full corpus, but absolute numbers will differ.
- **Single first stage**: we only rerank the dense top-100. Reranking BM25 top-100, or a hybrid BM25+dense fusion, is the natural next step.
- **Latency**: the cross-encoder is the slowest component in the pipeline by ~2 orders of magnitude. Production-realistic deployments would use a smaller distilled reranker or restrict rerank to top-50.

9. Next

- Rerank BM25 top-100 (W2 run) and compare the delta to the dense delta — does the reranker recover more from a weaker first stage?
- Hybrid first-stage (RRF over BM25 + dense) → rerank.
- Connect the reranked top-K to the W3 RAG generator and measure whether better ordering improves answer quality.